

Rank-nullity

$$T: V \longrightarrow W$$



two finite dimensional
F-vector spaces

Defⁿ

$$(1) \quad \ker(T) = \{v \in V : T(v) = 0\} \\ \parallel \\ \ker T \subseteq V$$

$$(2) \quad \operatorname{im}(T) = \{T(v) : v \in V\} \\ \subseteq W$$

Observation: (a) $\ker(T) \subseteq V$ is an
F-subspace

(b) $\operatorname{im}(T) \subseteq W$ is an
F-subspace

Pf: (a): (+) $T(v,1) = 0$ & $T(v,1) = 0$

$$\begin{aligned} \Rightarrow T(v_1 + v_2) &= T(v_1) + T(v_2) \\ &= 0 + 0 \\ &= 0 \quad \checkmark \end{aligned}$$

$$(\cdot) \quad T(v) = 0$$

$$\begin{aligned} \Rightarrow T(c \cdot v) &= c \cdot T(v) \\ &= c \cdot 0 \\ &= 0 \quad \checkmark \end{aligned}$$

$$(0) \quad T(0) = 0 \quad \checkmark$$

Proposition (Rank-nullity)

$$\underbrace{\dim \ker(T)}_{\text{null}(T)} + \underbrace{\dim \text{im}(T)}_{\text{rank}(T)} = \dim(V)$$

Pf.: Choose bases $\mathcal{B}_1$ for V

$\mathcal{B}_2$ for W

$$T \rightsquigarrow [T]_{\mathcal{B}_2, \mathcal{B}_1} \in M_{n \times m}(F)$$

$\underbrace{\hspace{10em}}_A$

$\underbrace{\hspace{10em}}_{\substack{n \text{ rows} \\ m \text{ columns}}}$

$$\beta_2 = (w_1, w_2, \dots, w_n)$$

$$n = \dim(W)$$

$$m = \dim(V)$$

Then to solve

$$T(x) = w$$

It is equivalent to solving

a matrix equation

$$[A | w']$$

Where $w' = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix} \in F^n$

$\parallel$
 $U_2^{-1}w$

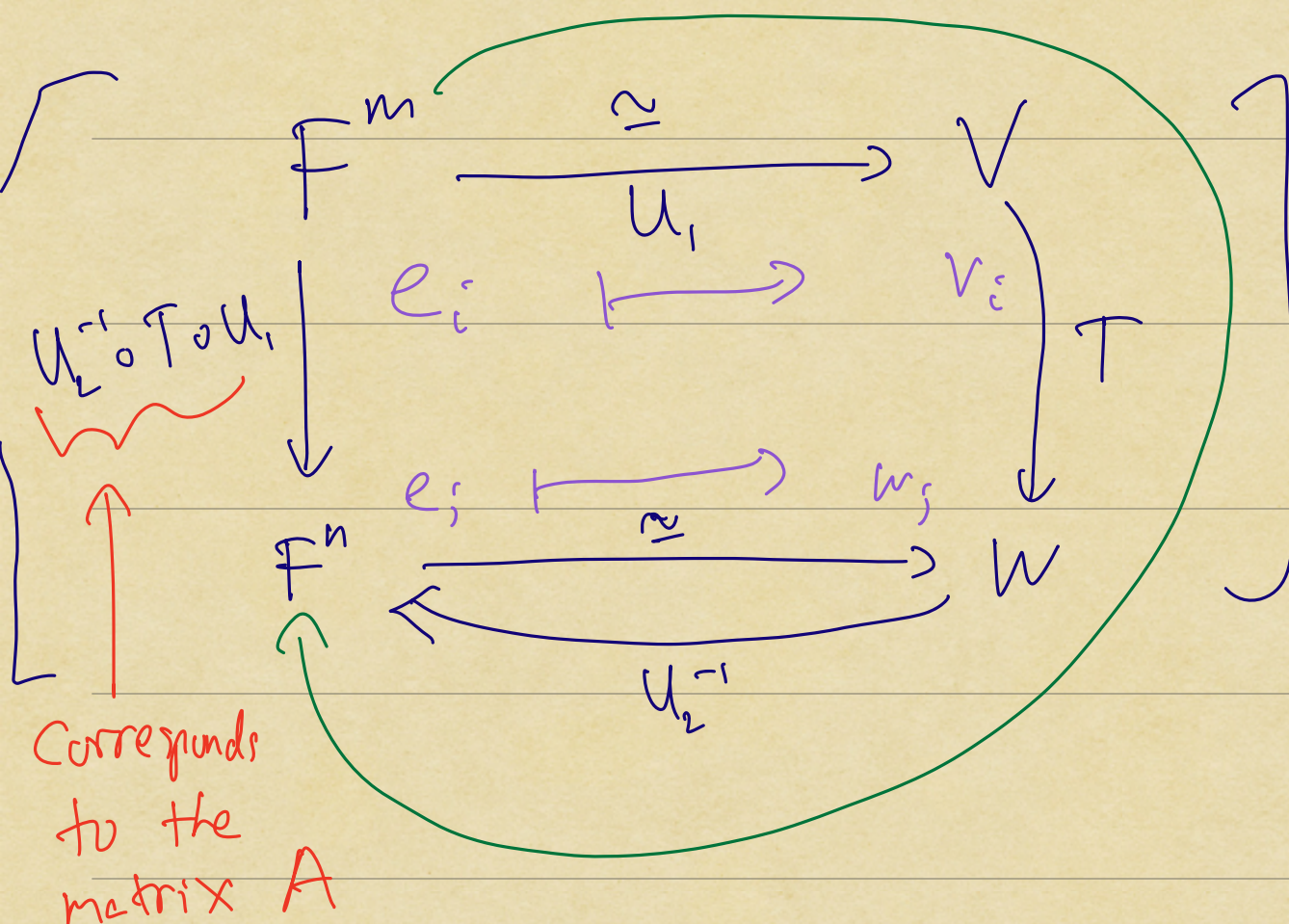
is determined by

$$w = a_1 w_1 + a_2 w_2 + \dots + a_n w_n$$

Why is this!

$$\beta_1 = (v_1, \dots, v_m)$$

$$\beta_2 = (w_1, \dots, w_n)$$



Solving $T(x) = w$, $x \in V$

$$\Leftrightarrow \underbrace{(u_2^{-1} T u_1)}_A (y) = u_2^{-1} w, y \in F^m$$

More precisely: If we solve for $y \in F^m$ then we recover x by setting it equal to $u_1(y)$

Concretely: If $y = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_m \end{bmatrix}$

then $x = h_1 v_1 + \dots + h_m v_m$

We are therefore solving for

$$Ay = U_2^{-1} w = w'$$

($\Rightarrow$) Solving

$$[A \mid w']$$

$$\underbrace{\quad}_{n \times m} \quad \underbrace{\quad}_{\eta \times 1}$$

$$\boxed{\dim \ker(T)}$$

We're solving for

$$T(x) = 0$$

$(\Leftrightarrow)$

$$[A | 0]$$

$$\boxed{\dim \operatorname{im}(T)}$$

For which $w' \in F^n$ can
we solve

$$[A | w'] ?$$

Process for both involves row reduction

Step (1)

Use row operations

(a) Swapping rows

(b) (non-zero) scalar multiplication

(c) Adding a multiple of one row to a different row

to get A to the form

leading 1s

$$\begin{bmatrix} 1 & a & 0 & c & 0 & d \\ 0 & 0 & 1 & b & 0 & e \\ 0 & 0 & 0 & 0 & 1 & f \\ 0 & 0 & 0 & 0 & 0 & g \\ 0 & 0 & 0 & 0 & 0 & h \end{bmatrix}$$

A is in reduced row echelon form

Features: (a) In every row, the non-zero entry is 1

(b) If a column has a leading 1, then all other entries in it are zero

(c) Leading 1s travel left to right

Step 2

free variables

$$\begin{bmatrix} 1 & a & 0 & b & 0 \\ 0 & 0 & 1 & c & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix}$$

$$x_1 + a x_2 + 0 \cdot x_3 + b x_4 = 0$$

$$x_3 + c x_4 = 0$$

x_2, x_4 can be arbitrarily valued
& x_1, x_3 are determined by these values

In general

$$\dim \ker(T) = \# \text{ of free variables}$$

$$\begin{bmatrix} 1 & a & 0 & b & 0 \\ 0 & 0 & 1 & c & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{matrix}$$

↑
given

need to solve

$$x_1 + a x_2 + 0 \cdot x_3 + b x_4 = y_1$$

$$x_3 + c \cdot x_4 = y_2$$

$$\begin{matrix} 0 = y_3 \\ 0 = y_4 \\ 0 = y_5 \end{matrix}$$

Equation can be solved $\Leftrightarrow$

$y_3 = 0$
 $y_4 = 0$
 $y_5 = 0$

$$\dim \operatorname{im}(T) = \# \text{ of leading } 1\text{'s}$$

Rank-nullity is now simply saying

of free variables

+

= # of columns

of leading 1's